Frontier



Unemit

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Unemit

Organization location (we welcome applicants from anywhere in the world)

Toronto, Canada

Name(s) of primary point(s) of contact for this application

Christian Haakonsen

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

For-profit startup developing technology for unemitting CO_2

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Between a slow rock and a hard place: a better air contactor to break out surficial mineralization

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

Surficial mineralization needs a better air contactor

The hard place of CDR is a tennis court, or more specifically that you need to touch all the atmosphere between it and space to remove a single ton of CO_2. There are plenty of rocks that can mineralize CO_2, but the abundant ones do so too slowly in air to just spread them out on that court. Unemit is working to break surficial mineralization out of this vise by developing a better gas-solid contacting approach for alkaline feedstocks.

A slow rock is manageable if you can accelerate mineralization by raising the temperature and CO_2 partial pressure, but that pesky hard place means we're stuck with the ambient CO_2 concentration and weather; the laws of thermodynamics crush any hope of heating or compressing all that air. So the obvious other thing to do is speed up the rock, and fortunately there's a growing body of work on increasing the reactivity of different feedstocks, laying out a menu of pretreatment options with different tradeoffs in terms of energy use and cost.

Such faster rock is still slow though, even with optimal weather. And the weather typically won't be optimal since the location will be constrained by regulations and access to feedstock, equipment, labor, and clean energy. That's true both near term, where the obvious place to start is with mine tailings, as well as longer term when scaling further through dedicated mining. Not only that, but certain conditions can lead to passivation of the feedstock that prematurely terminates mineralization, so the weather can be both a friend and a foe.

In other words, we're stuck needing to mineralize massive amounts of slow feedstock simultaneously and would like to pick the hours with helpful weather and avoid the harmful ones. That's the job of an air contactor, but unfortunately no one has solved how to make a suitable one yet.

It boils down to making massive cheap, yet efficient

Gigatons is a daunting scale, but one that mining already operates at. That's good, because for one gigaton of CO_2 we'll need at least several gigatons of feedstock. As an example, lizardite is a potentially attractive mineral for surficial mineralization that has enough magnesium to bind almost half a ton of CO_2 per ton of feedstock (0.5 t_{CO2}/t_F), though under ambient conditions the best case is only 80% of that and in practice significantly less. If we achieve, say, 40% in six months, which would be a stretch and take substantial pretreatments, then for a gigaton per year of carbon removal capacity we'd need almost three gigatons of lizardite pretreated and sitting in air contactors at any one time, requiring a massive amount of contactor volume.

Through that volume we need to flow about a billion tennis courts worth of atmosphere every year, intermittently based on the weather, which is a lot but doable with fans. While the fan efficiency matters, what really determines the energy needed to power them is the pressure drop through the air contactor and how much of the CO_2 it captures from the transiting air. For example, for an average fan efficiency of 50%, a 100 Pa pressure drop with 300 ppm CO_2 depletion requires 0.1 MWh/t_{CO2}.

That's fairly efficient air contacting, and the challenge is achieving that on the cheap.

Success is critical for the overall economics

The single most important factor in the economics of surficial mineralization is the achievable CO_2 capacity per ton of feedstock, which sets how much material must be obtained and flow through the system. The two things determining that for a given feedstock are the pretreatments and the reaction conditions in the air contactors. Both need to work well enough in concert to limit feedstock costs while being realized at sufficiently low cost themselves to not ruin the economics, as detailed in Fig. 1 below.

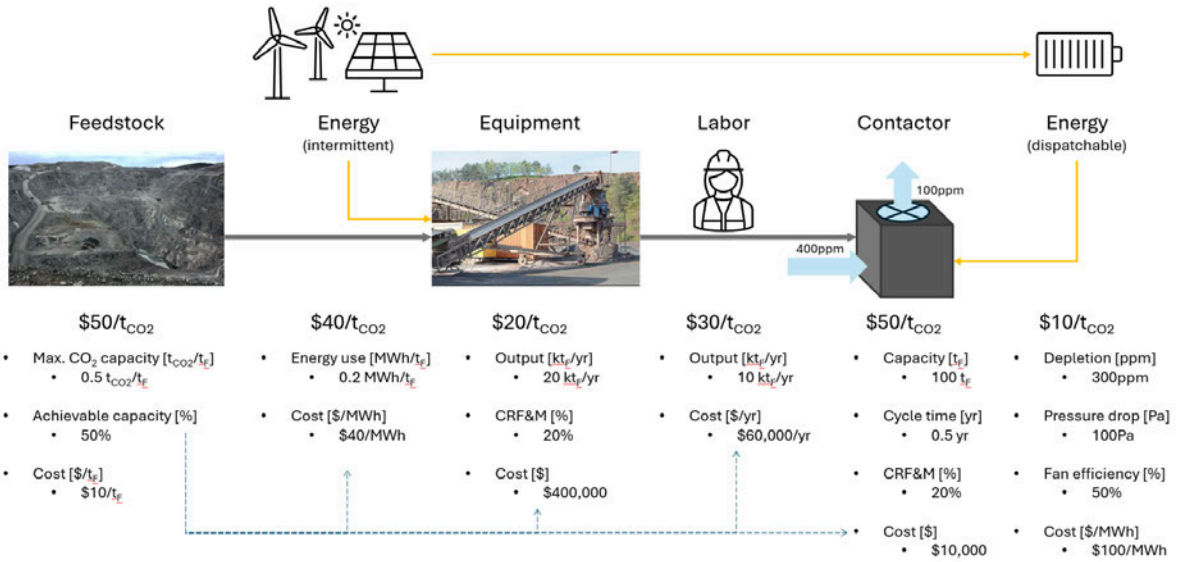


Figure 1: *Key technoeconomic drivers of surficial mineralization, with values showing one way to achieve ~$200/$t_{CO2}$; CRF&M is a combined capital recovery and maintenance factor. Different feedstocks and locations will have different optimal balances of feedstock, pretreatments, and air contacting, and these will evolve as costs fall with scale and experience.*

Pretreatments have a head start, also thanks to larger overlaps with methods in mining and other industries. The energy-intensive ones will benefit from being adapted to run intermittently to match the availability of cheap renewable energy, and can be pushed further in the future as the cost of such energy continues to fall.

That leaves air contacting as the most pressing technology gap. Its share of the final cost will be meaningful, but only because it has dramatically lowered the other cost contributions by shrinking the amount of feedstock passing through the system. It will have a steep experience curve, but riding it hinges on ensuring megaton-scale costs approach $200/$t_{CO2}$ to unlock sustainable demand. And so that is our research goal: demonstrate that our air contacting approach can enable surficial mineralization to reach $200/$t_{CO2}$ at scale.

Breaking out surficial mineralization starts the CDR flywheel

No CDR approach has yet managed to hit all the requirements for unlocking sustained exponential demand growth. Surficial mineralization is inherently durable, has intuitive and robust MRV, can start scaling quickly with existing mine tailings and keep up with growing demand, and naturally extends to near limitless scale with dedicated mining. Better air contacting is needed for the latter two to hold true, and to address the final major friction: cost. Compliance markets have shown that government-driven demand already can kick in around $\$200/t_{CO2}$, so the task at hand is to make the path there credible enough that forward-thinking voluntary buyers continue to help spin up the flywheel. Our air contactors will provide that path via surficial mineralization, making supply cheap and predictable enough to get governments comfortable dialing up demand. That demand predictability will carry with it other CDR approaches, many of which will be needed to clean up the mess we continue to make of the atmosphere.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Existing contacting approaches either don't scale or are too expensive

The best answer is often the simplest one, and in the case of air contacting the simplest approach would seem to be to spread the feedstock out on the ground, potentially with some additional farming-style manipulations. Unfortunately, while that may work for some high-reactivity feedstocks or to get started at a mining site, it doesn't really scale. Achieving a good capture rate with mine tailings still means needing a 200 m^2 footprint to capture one ton per year of CO_2, so a gigaton per year would require more than 10% of the arable land in the US or India. All that land is also needed close to suitable mines, and even where it could be available there'd be significant cost associated with transporting and distributing the feedstock. There would also be concerns around dust, both in terms of the cost and water use associated with mitigating it over such a large area, in addition to what would happen in a drought or other extreme weather exacerbated by climate change. So just spreading out the feedstock turns out to not be all that simple after all.

Another extreme is to shrink the footprint by going vertical, relying on automation to distribute and collect feedstock in giant warehouse-like fanned contactors. Robots are expensive though, and this approach can easily require more structure than it holds feedstock, so even for high-performing feedstock in a DAC loop the economics turn out to be challenging. For slower feedstocks they just don't work.

Swinging the pendulum back toward simplicity, another approach is to pile up the feedstock and blow air through it. The random and constricted paths available to the air lead to significant pressure drop per unit height of the pile, which means it must either be very shallow or large amounts of energy must be expended to force the air through it. That energy is needed when the ambient conditions are conducive to capture, which is not necessarily when electricity is cheap, so the cost quickly becomes prohibitive if the pressure drop exceeds a few hundred pascals. Meanwhile, limiting pressure drop with a shallow pile starts to pick up some of same challenges as just spreading the feedstock on the

ground, while also growing the area that needs to be fed with ducts from fans. Again, neither cheap nor as simple as the approach first appeared.

That leaves a gap in gas-solid air contacting between the deceptively simple approaches with larger footprints and the compact but too expensive DAC-style ones.

This project will unlock the golden middle ground

As shown in Fig. 2 below, we've discovered a way to unlock the golden middle ground where scalable but slower feedstocks can be harnessed in much simpler and cheaper structured contactors. Structuring enables fans to supply air to a large amount of feedstock with limited pressure drop, keeping energy costs in check while ensuring the ability to make optimal use of variable ambient conditions. It also significantly shrinks the footprint to enable reaching meaningful scale within existing mine sites, which together with the containment and control it affords will ease permitting and deployment.

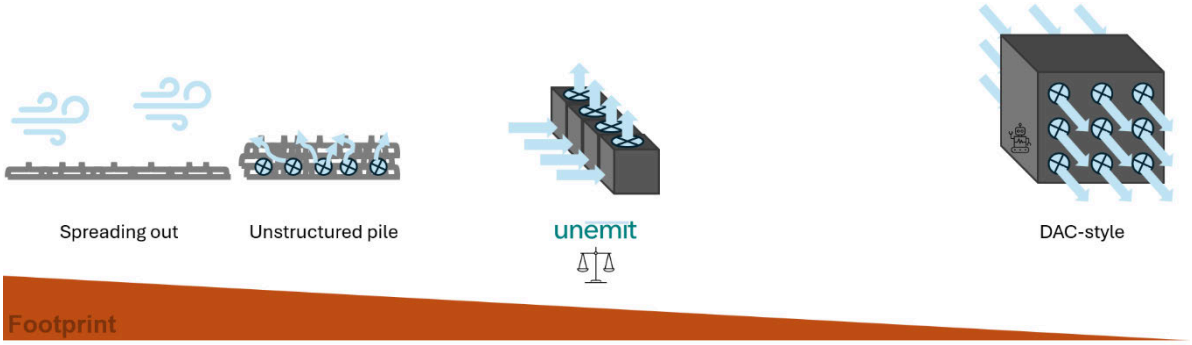


Figure 2: *Our air contacting approach relative to existing approaches. It strikes an ideal balance that significantly reduces footprint without adding cost-prohibitive complexity.*

This project will demonstrate that our novel air contacting approach is viable and can be made cheap enough to unlock massive scale. The best way to do that is to actually build a prototype, or rather a succession of them to retire key technical risks and show that a prototype with a full-scale air path can be built and operated with acceptable pressure drop. The key research question is then whether and how that can be done.

It's all 'bout the money (and scalability)

Cost is the ultimate metric, and it's easy to fall into the trap of optimizing for particular proxies or parts of the system without adequately considering the interconnected nature of the problem (cf. Fig. 1). We're squarely focused on expected large-scale technoeconomics, rapidly iterating between modeling, experiments, and actually building things to identify the most attractive parts of the option space and quickly run into any unknown unknowns. That means irreverently abandoning sunk cost, whether because something doesn't work well or because it turns out to be much easier than expected and so enables simplification.

Constraints drive suboptimal outcomes and amplify risks, so we pursue simplicity and decoupling wherever possible. Surficial mineralization is uniquely well-suited for decoupling since solids can be easily stored. Decades-old tailings piles that now become feedstocks are an extreme example, but between several key steps it can be helpful to maintain a buffer to ensure the operations of each can be optimized based on what matters most for that step, be it weather or energy cost. This also allows for less overengineering and redundancy, since interruption of any one step doesn't bring the whole process to a halt. Decoupling and simplification thus reinforce each other, and can be a powerful recipe for lowering cost.

Picking the right feedstock is key

Eventually, scalability will boil down to picking the right feedstocks and pretreatments. There may be others that are useful stepping stones early on, but in the end only choices that can get us past a gigaton per year will make enough of a dent. Because of the importance of achievable CO_2 capacity per ton feedstock in Fig. 1, those are likely to be magnesium and calcium silicate minerals. If there was abundant and pure portlandite or brucite easily available we'd be home free, but it seems we'll have to work a bit harder to source the alkalinity.

Wollastonite stands out as having comparable dissolution rates to brucite under ambient conditions, so would be very interesting if there was more of it accessible. Alas, true scalability likely rests on abundant mafic and ultramafic rocks with dissolution rates several orders of magnitude lower. The good news is that there are lots of those lying around as nickel and asbestos mine tailings, already extracted and partially pretreated, and we know how to get more when we need to.

While such mine tailings often contain some brucite, which can be carbonated more easily, it's too small a fraction for it to make sense to mineralize the feedstock directly in an air contactor. However, it could be leached, along with more magnesium from the silicates, to provide brucite for our air contactors. Advancements in alkalinity extraction may make that an attractive option, though our current ambition is to instead directly carbonate a larger share of the feedstock. That will require substantial additional pretreatment, likely including comminution to particle sizes below 10 microns and/or dehydroxylation; both energy-intensive processes but ones amenable to intermittent operation and so low and falling cost of renewable energy.

In parallel with air contactor prototyping, this project will test various feedstocks and pretreatments under simulated operational conditions in bench-scale experiments, measuring carbonation rate and extent and characterizing diffusive CO_2 transport. Moisture plays a key role in dissolution, reaction, and precipitation, so weather data for potential sources of feedstock will be analyzed to inform the testing. The results will guide feedstock and pretreatment choice, inform air contactor design parameters and operational strategies, and be used to predict overall performance and economics; once we've answered the question of which feedstocks and pretreatments are most promising.

Time is of the essence

The sooner we can show that surficial mineralization can be done cheaply and at scale, the sooner we can get on our way to gigatons of CDR. This project builds on all the great work that has been done to date, adding the last piece of the puzzle that finally brings it all together.

d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

Category	*Risk*	*Mitigation*
Scientific	Poor feedstock performance	Run experiments in parallel, monitoring and branching or starting additional ones based on developments; apply more aggressive pretreatments
Technical	Difficulty scaling air contactor	Rapidly prototype while attempting to anticipate how things will scale, and swiftly get to actually trying them
	Insufficient dust control	Measure dust and explore strategies to minimize using moisture and fan control
Project execution	Inability to access feedstock	Limit amount required for initial testing, identify commercial versions or approximations, and seek out partners who can provide
Financial	Drop in currency exchange rate	Front-load grant payments and exchange promptly

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables *(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)*	Determination of suitability of one or more promising feedstocks with air contacting approach Pressure-drop measurements from a contactor prototype with full-scale air path TEA based on measured feedstock carbonation rate and extent
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR*	Surficial mineralization bottlenecks resolved: land use, dust, feedstock

field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)	scalability, and cost
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	Q1 2027: Test compatibility of promising feedstock with contacting approach Q4 2027: Measure pressure drop of contactor prototype with full-scale air path Q2 2028: Model technoeconomics based on measured feedstock performance
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$750k